

Course overview “Theoretical Ecotoxicology: Concepts and Application”

University Duisburg-Essen, 2026

Lecturers: Ralf B. Schäfer, Verena Schreiner, Hanh Hong Nguyen, Helena Bayat, Paul Ozoh, Iris Pimentel, André Gergs, Marie Brasseur, Silviu Bobric

The course consists of a lecture with practical exercises at the computer and typically runs from 14:00-17:15 with a 15 minute break in the room S06 S00 B08. The dates and topics are as follows:

Date	Topic
15.4.2026	General introduction (including empirical vs. mechanistic model types), useful tools (e.g. Quarto/Markdown) and software installation. Teaser projects
22.4.2026	Ecotoxicological databases and automated access of databases
29.4.2026	Toxicity prediction
06.5.2026	Dose-response modelling
13.5.2026	Mixture modelling
20.5.2025	Multiple stressor modelling
27.5.2026	Toxicogenomics modelling I
03.6.2026	Toxicogenomics modelling II
10.6.2026	Dealing with complex data and designs (incl. Mixed effect models)
17.6.2026	Bioaccumulation modelling
24.6.2026	TKTD modelling I
01.7.2026	TKTD modelling II
08.7.2026	Introduction ecological modelling I
15.7.2026	Introduction ecological modelling II
22.7.2026	Spatial analysis and Introduction projects and portfolios